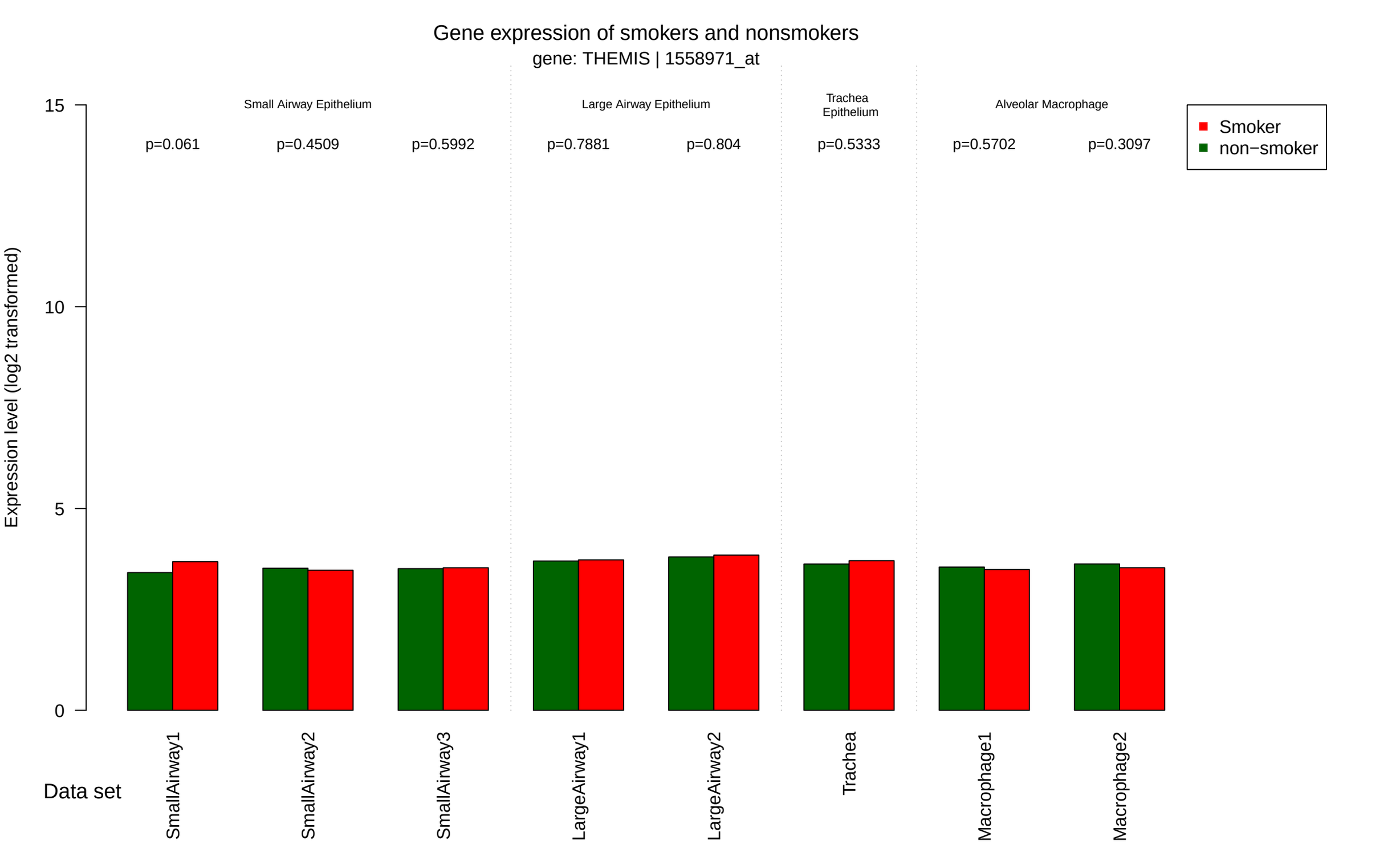
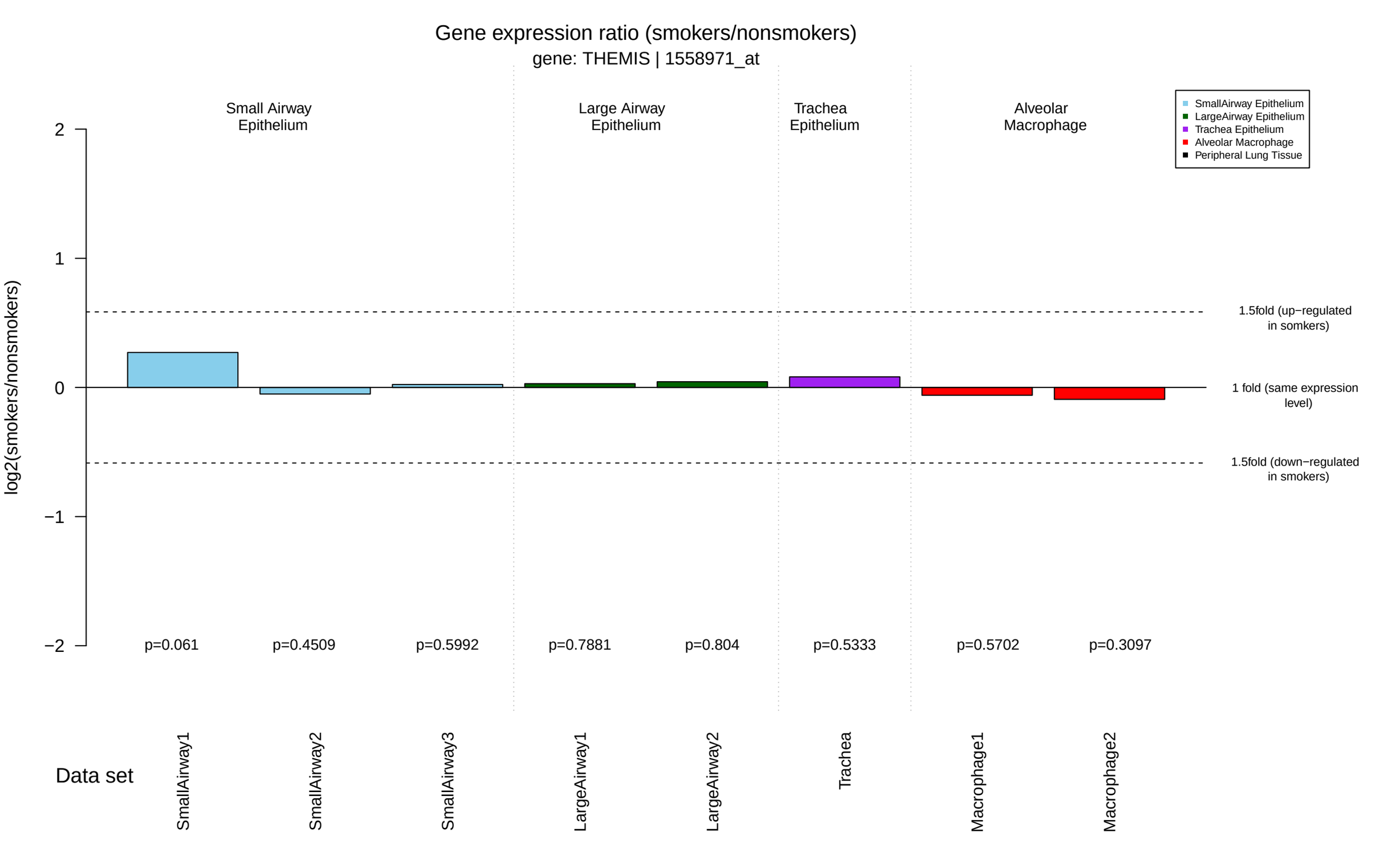
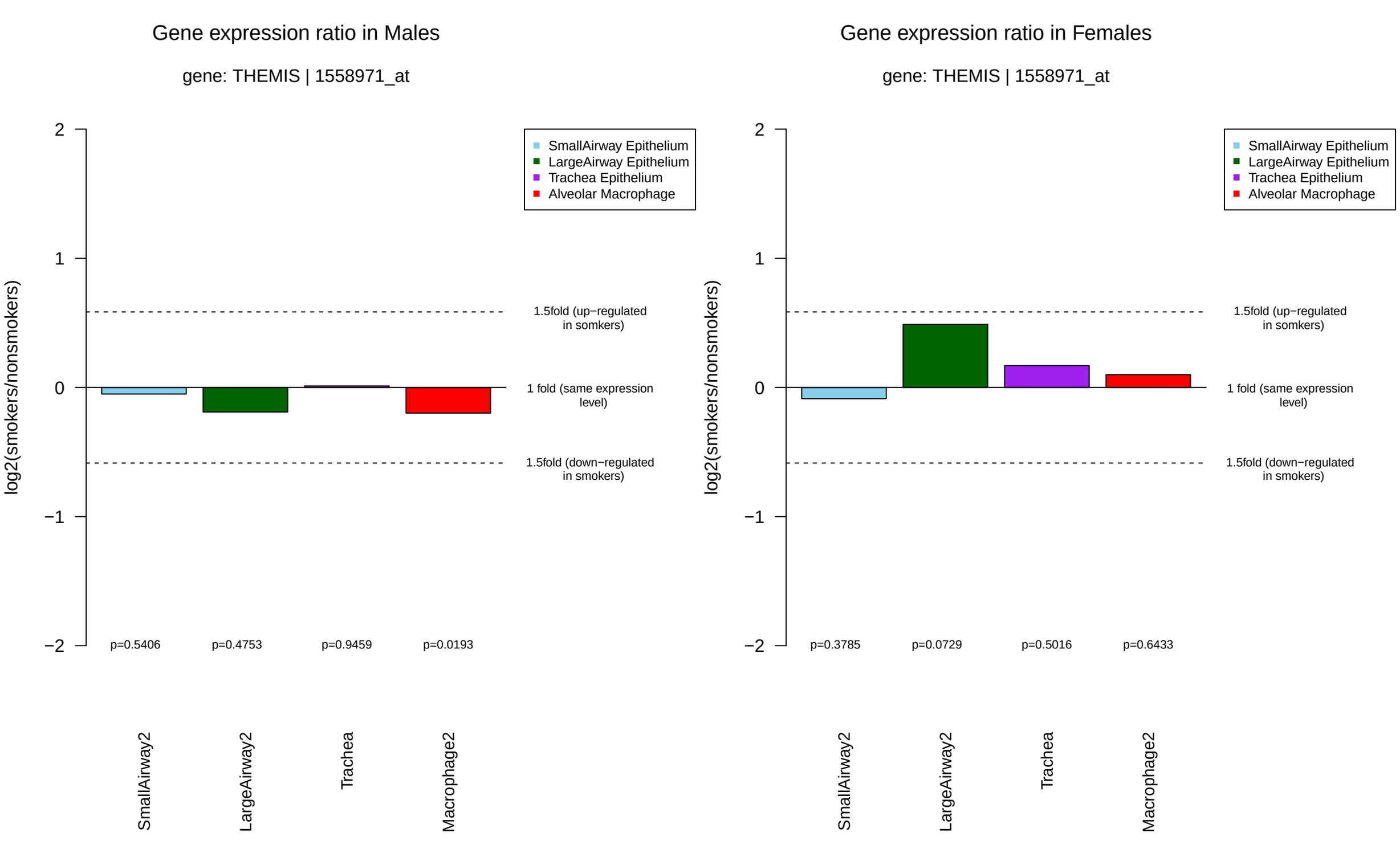
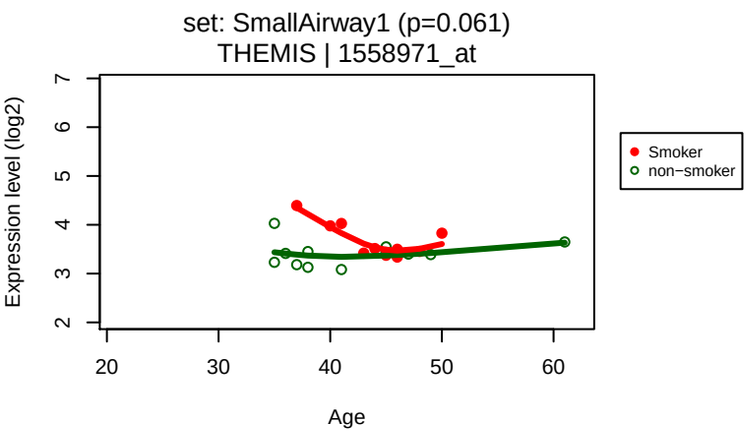
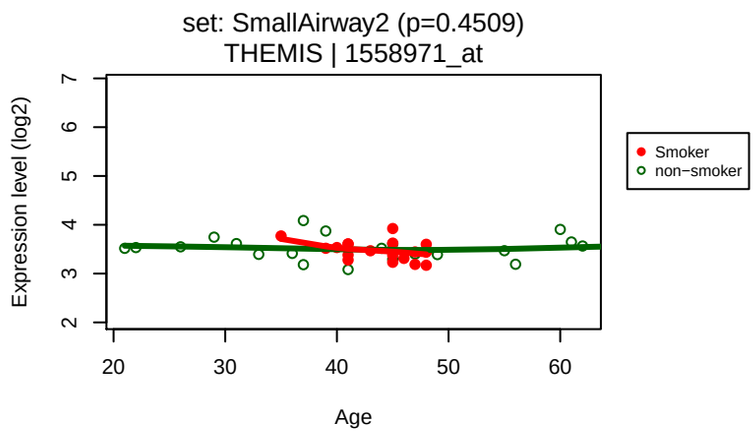
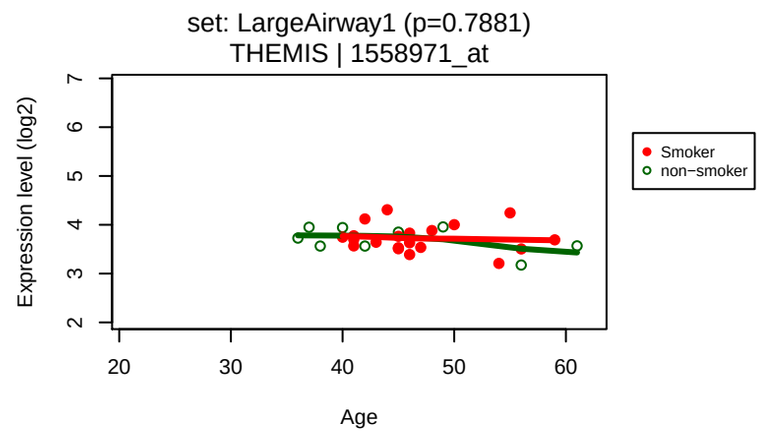

Gene expression of smokers and nonsmokers
gene: THEMIS | 1558971_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.061
p=0.4509
p=0.5992
p=0.7881
p=0.804
p=0.5333
p=0.5702
p=0.3097
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

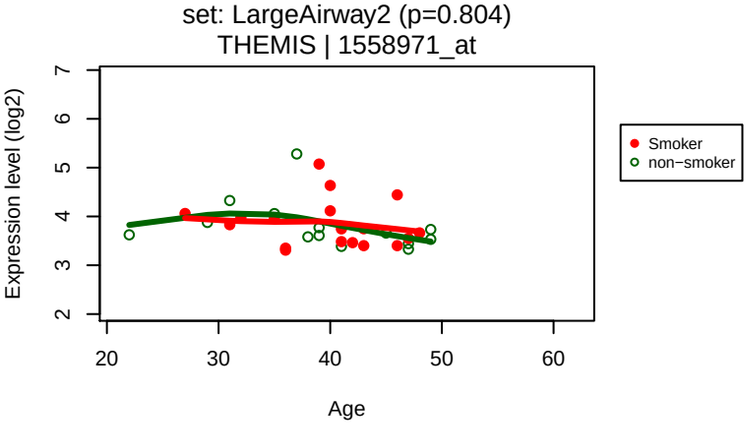

Gene expression ratio (smokers/nonsmokers)
gene: THEMIS | 1558971_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.061
p=0.4509
p=0.5992
p=0.7881
p=0.804
p=0.5333
p=0.5702
p=0.3097
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

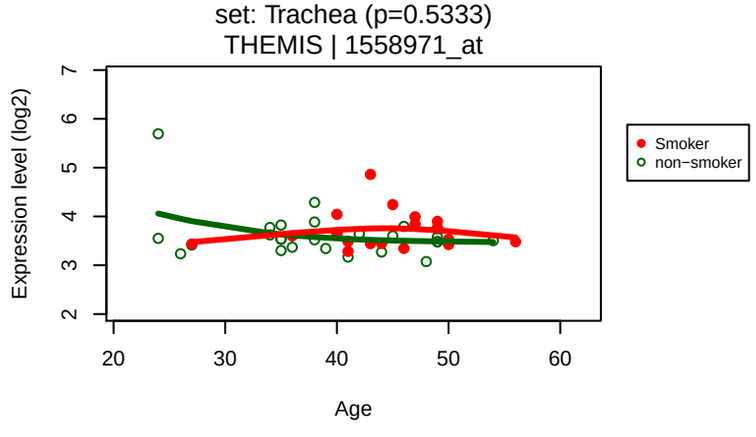

Gene expression ratio in Males
gene: THEMIS | 1558971_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5406
p=0.4753
p=0.9459
p=0.0193
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: THEMIS | 1558971_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3785
p=0.0729
p=0.5016
p=0.6433
SmallAirway2
LargeAirway2
Trachea
Macrophage2

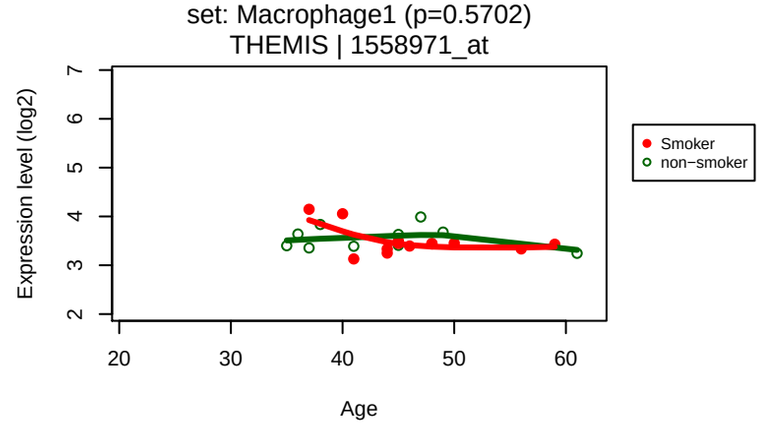

set: SmallAirway1 (p=0.061)
THEMIS | 1558971_at
Expression level (log2)
Age
Smoker
non−smoker

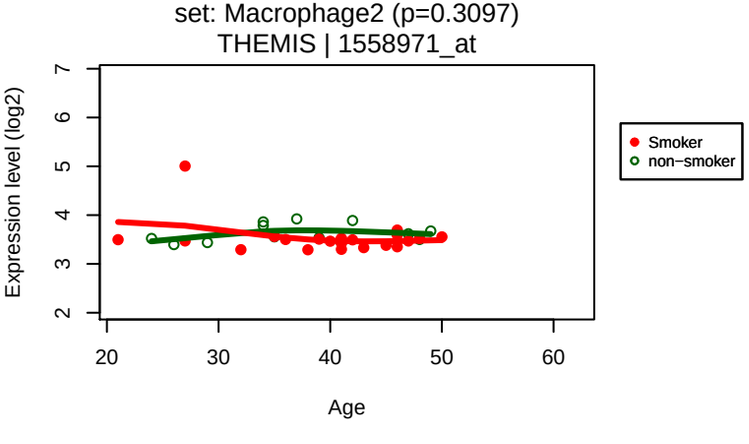

set: SmallAirway2 (p=0.4509)
THEMIS | 1558971_at
Expression level (log2)
Age
Smoker
non−smoker

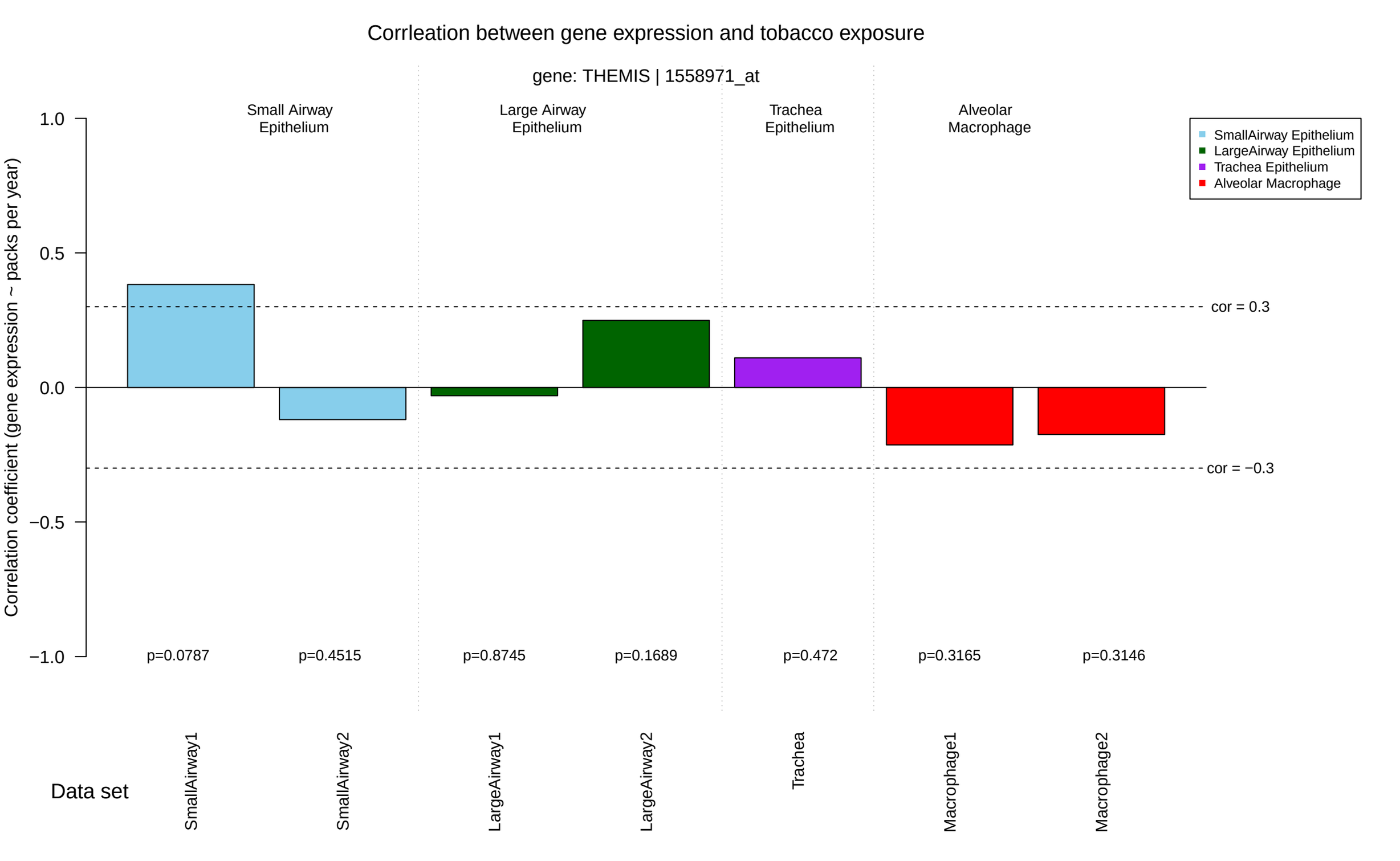

set: LargeAirway1 (p=0.7881)
THEMIS | 1558971_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.804)
THEMIS | 1558971_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5333)
THEMIS | 1558971_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5702)
THEMIS | 1558971_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3097)
THEMIS | 1558971_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: THEMIS | 1558971_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0787
p=0.4515
p=0.8745
p=0.1689
p=0.472
p=0.3165
p=0.3146
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2